THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

032/1 CHEMISTRY 1

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TIME: 3:00 HOURS YEAR: 2025

INSTRUCTIONS

1. This paper consists of section A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and **two (2)** question from section C.
3. Section A carries **sixteen (16)** marks, section B carries **fifty-four (54)** marks and section C carries **thirty (30)** marks.
4. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
5. All communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer sheets provided.
7. Where necessary, the following constants may be used
   - Atomic masses: H = 1, C = 12, O = 16, CL = 35.5, Na = 23, Zn =65, Ca = 40, Cu=64, S = 32,
   - GMV at S.T.P = 22.4 dm^3
   - 1 faraday = 9650 coulombs

SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x) choose the correct answer from the given alternative and write its letter besides the item number in the answer booklet provided.

(i) Which is the correct description of nucleons?
- A Nucleons are neutrons and protons in the nucleus of an atom
- B Nucleons are neutrons in the nucleus of an atom
- C Nucleons are protons and electrons in the nucleus of an atom
- D Nucleons are neutrons and electrons in the nucleus of an atom
- E Nucleons are neutrons, protons and electrons in the nucleus of an atom.

(ii) Which one the following characteristics ions formation?
- A Metal atoms gaining electrons in their outermost shells
- B Non- metal atoms losing electrons from their outermost shells
- C Metal atoms losing electrons from their innermost shells
- D Non-metal atoms gaining electrons in their innermost shells
- E Metal atoms losing electrons from their outermost shells

(iii) The following are steps of writing ionic equation except
- A Writing the correct formula for the reaction
- B Writing all soluble ionic substances
- C Writing the reactions in words
- D Writing all insoluble ionic products
- E Writing balanced formula equation for the reaction

(iv) A form Two student was given the following staffs for preparation of Oxygen gas.
- (i) Source of heat
- (ii) Manganese dioxide
- (iii) Hydrogen peroxide
- (iv) Potassium chlorate

Which combination will fast produce oxygen?
- A (ii) and (iv)
- B (i) and (iii)
- C (iii) and (iv)
- D (i). (iii) and (iv)
- E (i), (ii) and (iv)

(v) Consider the given trends in physical properties of elements in the periodic Table.
- (i) Electron affinity increase from left to right
- (ii) Densities increases down the group
- (iii) Melting point of metals increases down the group
- (iv) Metallic character increases from left to right

Which combination demonstrates correct trend?
- A (iii) and (iv)
- B (i) and (iii)
- C (ii) and (iii)
- D (i) and (ii)
- E (i) and (iv)

(vi) The organic compounds marked **A** and **B** reacted together to form compound **C** as shown in the equation:

$$\underset{\mathbf{A}}{CH_3CH_2OH_{(1)}} + \underset{\mathbf{B}}{CH_3COOH_{(l)}} \xrightarrow{Conc.H_2S0_4} \underset{\mathbf{C}}{CH_3COOCH_2CH_{3(1)}}$$

What are the names of compounds A, B and C?

A Ester, ethanoic acid and alcohol
B Alcohol, carboxylic acid and ester
C Alcohol, ester and carboxylic acid
D Ethanol, ethanoic acid and ethyl ethanoate
E Carboxylic acid, ethanol and ethyl ethanoate

(vii) Which of the following does not constitute one mole?
A 32g of Oxygen molecules
B 2g of hydrogen molecules
C 19g of hydroxonium ions
D 48g of carbon monoxide molecules
E 98g of sulphuric acid

(viii) When a student mixed a solution of certain copper (ii) salt and sodium hydroxide, blue precipitate is formed. Which substance was produced?
A Copper (II) Oxide
B Copper hydroxide
C Copper salt
D Copper (I) Oxide
E Copper metal

(ix) Permanent hardness of water can be removed by using washing soda. Which reaction is the correct ionic equation for the softening process?
A $Ca^{2+}_{(aq)} + S04^{2-}_{(aq)} \longrightarrow CaS0_{4(aq)}$
B $Ca^{2+}_{(aq)} + CO^{32-}_{(aq)} \longrightarrow CaCO_{3(s)}$
C $Mg^{2+}_{(aq)} + S0_4^{2-}{}_{(aq)} \longrightarrow MgS0_{4\,(aq)}$
D $Ca^{2+}_{(aq)} + 2HCO^{-}_{3(aq)} \longrightarrow Ca(HC0_3)_2(aq)$
E $Mg^{2+}_{(aq)} + 2HCO_{3\,(aq)} \longrightarrow Mg(HCO_3)_{2(aq)}$

(x) Why sodium hydroxide pellets should be stored in a closed container?
A Sodium hydroxide is efflorescence
B Sodium hydroxide is hygroscopic
C Sodium hydroxide is deliquescent
D Sodium hydroxide is volatile
E Sodium hydroxide is flammable.

2. Match the description of organic compound in **List A** with the corresponding response in **List B** which are organic terms by writing the letter of the correct response beside the item number in the answer sheet provided

LIST A	**LIST B**
(i) Existence of two or more compounds having the same molecular formula but different structural formula (ii) Example of Unsaturated hydrocarbon (iii) Any atom or group of atoms, except hydrogen which is bonded to the longest continuous carbon chain. (iv) Compounds of carbon and hydrogen. (v) An arrangement of the members of the family of organic compounds in order of increasing molecular masses (vi) Functional group of alkynes	A Alkene B Isomerism C Homologous series D Functional group E hydrocarbons. F Triple bonds G Isomer H Substituent

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) Sodium chloride and hydrogen chloride are both compounds of chlorine. Give Five (5) differences between the two compounds.

   (b) Write down the names of the following compounds:

   (i) MnO_2

   (ii) Al_2O_3

   (iii) N_2O_5

   (iv) $H_2C_2O_4$

4. Charcoal and firewood are used by most people in third world countries than gas fuel and electricity despite the fact that they are better sources of fuels. Using six (6) points justify the statement.

5. In practice, the speed of reaction is affected by concentration of reactants, surface area of solids and temperature. Explain how each of them affects the speed of reaction.

6. (a) During electrolysis, the selective discharge of ions depends on position of the electrochemical series, nature of electrodes and concentration of the competing ions. Explain how each of those affects the discharge of ions at the electrode.

(b) Find the mass of copper which will be deposited in electrolysis when a current of 0.45A is supplied for 25 minutes.

7. Students tested four sample of water each $10cm^3$ from different areas at Chunya District by shaking with three drops of soap solution. The observation were recorded as shown in the table below

Sample	Observation with soap solution	Observation for boiled sample with soap solution
A	No lather	Lather
B	Lather	Lather
C	Lather	Lather
D	No lather	No lather

(a) Which sample contain only temporary water? Give reason

(b) Which sample contain permanent hard water? Give reason

(c) Name the chemical substances that would be the cause of hardness water in sample mentioned in 7(a) and 7(b)

(d) Write the chemical equation for removing hardens of water in sample identified in 7(a) and 7(b)

8. (a) A form one student suggested that luminous flame and non-luminous flame are similar. As a knowledgeable form four student argue against by giving four (4) points.

(b) Halima was preparing food for her family using hot oil in a frying pan. Accidentally the pan tipped over and a huge fire spread on her kitchen floor.

(i) Which extinguishers would be appropriate for putting out the fire?

(ii) Which extinguisher would be dangerous to use to try to put out the fire?

(iii) In which class does a type of fire belong?

(c) A carpenter stored some iron nails in an open container after three days the nails turned reddish brown in colour.

(i) Why the colour of the nails changed to reddish brown

(ii) Outline two condition necessary for such changes to take place.

SECTION C (30 Marks)

Answer only **two (2)** questions from this section

9. What are the Five (5) immediate efforts to be done by National Environment Management Council (NEMC) in Tanzania in order to ensure that land is not polluted?

10. Sulphuric acid is very useful in daily life as it is used to make many items for example car batteries and fertilizers. The acid is manufactured in industries; Explain this process step-by step.

11. When dilute nitric acid is added to green solid **A**, a blue solution **B** is formed and gas **C** that forms a white precipitate with lime water. Heating the blue solution of the hydrated salt **B**, it initially loses its water of crystallization and the decompose to form a black solid **E**, brown fumes **F** and a gas **G** that relight a glowing splint.

    (a) Identify the solid **B** and gases **F** and **G**

    (b) Write the balanced chemical equation for the reaction between solid **A** and dilute nitric acid

    (c) Give a balanced chemical equation for the formation of solid **E**

    (d) Describe two chemicals' properties of gas **G**

    (e) Give the IUPAC name of green solid **A**